



COMSATS University Islamabad

Department of Computer Science

Course Syllabus

Course Information

Course Code: **CSC325**

Credit Hours: **3(2, 1)**

Lab Hours/Week: **3**

Course Title: **Computer Organization & Assembly Language**

Lecture Hours/Week: **2**

Pre-Requisites: **EEE240-Fundamental of Digital Logic Design**

Catalogue Description:

This course covers the concepts of computer organization along with programming in Assembly language. Topics include: Preliminary Concepts of Computer Organization; Von Neumann Architecture; IA-32 Microprocessors Organization; Modes of the processors; Non-Von Neumann Architectures; Pipelined vs. NonPipelined Systems; Parallel Processing; CISC vs. RISC Processors; Instructions Set Design & Formats; Addressing Modes; Memory System Organization & Architecture; Utilization of Memory Systems in Programming; IO Fundamentals; Data Transfer methods; CPU Performance Calculation; and Assembly Language Programming.

Text and Reference Books

Textbook:

1. Assembly Language for x86 Processors, Irvine, K.R., Pearson, 2020.

Reference Books:

1. Computer Organization and Architecture, William S., Pearson, 2019.
2. Essentials of Computer Organization and Architecture, Null L., Jones and Bartlett, 2019.

Week wise Plan:

Lecture #	CDF Unit #	Topics Covered	Reading Material
1.	1	Basic Computer Components, Computer Organization	Kip R:Ch1
2.	1	Von Neumann Architecture and Model, Instruction Execution Cycle, Instruction Set Architecture, Instruction Processing	Kip R:Ch1
3.	1	Modes of Operation/Processor; Registers; Real and Protected Address modes	William:Ch3
4.	1	X86 and X64 bit Processors	Kip R:Ch2
5.	2	Non-Von Neumann Architecture	William:Ch10,11
6.	2	Pipelined and Non-Pipelined Computer Systems, Pipelined System Clock Cycle, Efficiency. Superscalar Architecture.	William:Ch11
7.	2	Vector Processors and Parallel Processing. Comparison of CISC and RISC Processors.	William:Ch11
8.	2	Instruction set Format and Designs, Instruction format trade-offs	Kip R:Ch3
9.	2	Common Addressing Modes, Basic Addressing Modes, Register Addressing, X86 Addressing Modes	Kip R:Ch4

10.	3	Computer Memory System Organization, Memory Types i.e., Registers, Cache Memory, Main Memory, Secondary Memory	Kip R:Ch4
11.	3	Memory Hierarchy Insights, Locality Assignment No. 01 of Reference, Cache Mapping	Kip R:Ch4

12.	3	Direct Mapping, Fully Associative Mapping, Set Associative Mapping and PA(Physical Address)	William:Ch7
13.	4	Link Library Overview; Calling, Linking Book Library; Info to Library procedures	Kip R:Ch5
14.	4	User define Procedures, Program design using Procedures	Kip R:Ch5
15.	4	Boolean and Comparison Instructions (AND, OR, NOT, XOR, TEST, CMP)	Kip R:Ch6
16.	5	CPU Performance Calculations, Cycles Per Instruction (CPI) Cycles Per Instruction (CPI), CPU clock rate, Program Execution Time, CPU Average CPI	Kip R:Ch6
17.	Mid Term Exam		
18.			
19.	5	Factors Affecting CPU Performance, Aspects of CPU Execution Time Aspects of CPU Execution Time, Instruction Types & CPI, Frequency & CPI Instruction Frequency & CPI	Kip R:Ch6
20.	5	MIPS, Megaflops, and SPEC.	Kip R:Ch6
21.	6	Assembly Language Programming	Kip R:Ch4
22.	6	Basic Elements, Program Development Cycle	Kip R:Ch4
23.	6	Instruction Formats, Data Declaration & Addressing Modes	Kip R:Ch4
24.	6	Arrays & Structures Access	Kip R:Ch6
25.	6	Conditional and Unconditional Jumps	Kip R:Ch6
26.	6	Irvine Library and its Functions	Kip R:Ch10
27.	7	Assembly Language Programming Advanced Concepts	Kip R:Ch10
28.	7	Arithmetic Expressions	Kip R:Ch6
29.	7	Logical Operations, Conditional & Repetitions	Kip R:Ch6
30.	7	Procedures in Assembly Language and Macros	Kip R:Ch5
31.	7	Windows Programming	Kip R:Ch5
32.	7	High-level Language Interfaces	Kip R:Ch5
Final Term Exam			

Graduate Attributes (GAs)

S.#	Description
2	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.
3	Identify, formulate, research literature, and solve <i>complex</i> computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
4	Design and evaluate solutions for <i>complex</i> computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.

Course Learning Outcomes (CLOs)						
Sr.#	Unit#	Course Learning Outcomes	Blooms Taxonomy Learning Level	GA		
CLO's for Theory						
CLO-1	1-2	Articulate the components of a computer along with its architecture.	Understanding	2		
CLO-2	3-4	Review memory systems and its hierarchy along with IO methods.	Understanding	2		
CLO-3	5	Compute the performance of CPU.	Applying	2		
CLO-4	6-7	Demonstrate assembly language programming capabilities.	Applying	2,3		
CLO's for LAB						
CLO-5	1-4	Apply concepts of computer organization using a debugging tool.	Applying	2		
CLO-6	6-7	Construct assembly language programs of reasonable complexity.	Creating	2-4		
CLO Assessment Mechanism						
Assessment Tools	CLO-1	CLO-2	CLO-3	CLO-4	CLO-5	CLO-6
Quizzes	Quiz 1	Quiz 2	Quiz 3	Quiz 4	-	-
Assignments	-	Assignment 1	Assignment 2&3	Assignment 4	LAB Assignments	LAB Assignments
Mid Term Exam	Mid Term Exam	Mid Term Exam	-	-	Lab Mid Term Exam	-
Final Term Exam	Final Term Exam				-	Lab Project/ Lab Final Term Exam
Policy & Procedures						

- **Attendance Policy:** Every student must attend 80% of the lectures as well as laboratory in this course. The students falling short of required percentage of attendance of lectures/laboratory work, is not allowed to appear in the terminal examination.

- **Course Assessment:**

	Quizzes	Assignments	Mid Term Exam	Terminal Exam	Final Marks
Theory (T)	15	10	25	50	100

- **Grading Policy:** The minimum passing marks for each course is 50% (In case of LAB; in addition to theory, student is also required to obtain 50% marks in the lab to pass the course). The correspondence between letter grades, credit points, and percentage marks at CUI is as follows:

Grade	A	A-	B+	B	B-	C+	C	C-	D+	D	F
Marks	>= 85	80 - 84	75 - 79	71 - 74	68 - 70	64 - 67	61 - 63	58 - 60	54 - 57	50-53	< 50
Cr. Point	3.67-4.00	3.34-3.66	3.01-3.33	2.67-3.00	2.34-2.66	2.01-2.33	1.67-2.00	1.31-1.66	1.01-1.30	0.10-1.00	0.00

- **Missing Exam:** No makeup exam will be given for final exam under any circumstance. When a student misses the mid-term exam for a legitimate reason (such as medical emergencies), his grade for this exam will be determined based on the Department policy. Further, the student must provide an official excuse within one week of the missed exam.
- **Academic Integrity:** All CUI policies regarding ethics apply to this course. The students are advised to discuss their grievances/problems with their counsellors or course instructor in a respectful manner.
- **Plagiarism Policy:** Plagiarism, copying and any other dishonest behaviour is prohibited by the rules and regulations of CUI. Violators will face serious consequences.